

Infosys Internship — Individual Contribution Documentation

Milestone 3: Week 5 & 6 — Treatment Effectiveness Analysis & Healthcare Analytics

Contributor: Penchala Prasad | Project: St. Jude Medical Center — HealthForecast AI | Milestone: 3

1. Executive Summary & Overview

During Milestone 3 (Week 5 & 6), the project focused on implementing treatment evaluation workflows, pharmaceutical outcome analysis modules, hospital-wide performance dashboards, and healthcare analytics reporting systems.

2. Key Modules & Individual Contribution

As an individual contributor, I designed, built, and verified the following healthcare analytics modules:

- **Treatment Evaluation Workflows:** Implemented longitudinal recovery progress trajectories across Cardiac, Renal, and Respiratory specialties.
- **Medication Outcome Analysis Modules:** Created drug protocol efficacy vs side effect rate bar visualizers and a filterable medication outcome matrix with adherence ratings.
- **Executive Hospital Performance Dashboards:** Built hospital-wide recovery index, therapeutic efficacy %, complication prevalence tracking, and benchmark rating (4.8/5).
- **Departmental Benchmark & Outcome Analytics:** Integrated recovery rate vs readmission comparison matrices across Cardiology, Endocrinology, Pulmonology, and General Medicine.
- **Downloadable Patient Outcome Reports:** Implemented CSV report generation for executive clinical outcome evaluation.
- **Healthcare Trend Monitoring Tools:** Built population health incidence monitors and age-group demographic trend matrices.

■ Milestone 3 Outcomes Verification Matrix

Required Task / Deliverable	Target Outcome	Implementation Component	Status
Treatment evaluation workflows	Specialty recovery trajectory charts	/doctor/treatment-effectiveness	COMPLETED
Medication outcome analysis	Drug efficacy vs complication matrix	/doctor/treatment-effectiveness	COMPLETED
Healthcare performance dashboards	Hospital recovery index & 4.8/5 rating	/hospital-admin/analytics	COMPLETED
Patient outcome analytics reports	Department recovery vs readmission table	/hospital-admin/analytics	COMPLETED
Healthcare trend monitoring tools	Population health incidence monitors	/researcher/population-health	COMPLETED
Outcome report exporter	Downloadable patient outcome CSV reports	/hospital-admin/reports	COMPLETED

